

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

ORDER NO. 5-01-112

WASTE DISCHARGE REQUIREMENTS
FOR
ESPARTO WASTEWATER TREATMENT FACILITY

ESPARTO COMMUNITY SERVICES DISTRICT
YOLO COUNTY

The California Regional Water Quality Control Board, Central Valley Region (hereafter Board), finds that:

1. On 22 September 1989, the Board adopted Waste Discharge Requirements (WDRs) Order No. 89-165, prescribing requirements for the operation of a domestic wastewater treatment facility (WWTF) consisting of oxidation ponds for the Esparto Community Services District (CSD). Order No. 89-165 is neither adequate nor consistent with current plans and policies of the Board.
2. On 24 April 2000, a Report of Waste Discharge (RWD) was submitted by the Esparto CSD (hereafter Discharger) for the proposed expansion in the domestic WWTF. The RWD was found to be incomplete due to inadequate information on pond storage capacity. Two amendments were submitted to complete the RWD, dated 7 August and 28 August 2000, but these documents were also incomplete as a hydraulic balance for full expansion of the WWTF were not submitted. By letter dated 4 October 2000, Board staff indicated that the WDRs would be revised with provisions for submittal of the necessary documents.
3. The WWTF occupies 24.75 acres (Assessor's Parcel No. 49-130-07) on property owned by the Discharger. In February 2000, the CSD purchased 60 acres immediately east of the existing site to accommodate the expansion of the WWTF (Assessor's Parcel No. 49-120-23). The site is approximately 0.5 miles east of the town of Esparto on the north side of Highway 16, in T10N, R1W, MDB&M. The WWTF is approximately 1.7 miles south of Cache Creek and is bordered on the south by Lamb Valley Slough, and by South Fork Willow Slough on the east, as shown on Attachments A and B and part of this Order by reference.
4. The Esparto WWTF has been operating since 1963. Originally, wastewater was discharged to Lamb Slough under a National Pollutant Discharge Elimination System permit. In 1980, WDRs were issued for the discharge of wastewater to land with reclamation of the wastewater on approximately 20 acres to the southwest of the existing site. This area is owned by Bill Traylor and has been used primarily for the irrigation of pasture.
5. The existing WWTF services a few dozen small businesses, public schools, and 540 homes, and consists of a gravity feed conveyance system, a wet well/lift station and comminutor at the headworks, and six oxidation ponds (Nos. 1 to 6), as shown on Attachment C and part of this Order by reference. The comminutor is rated at 300,000 gallons per day (gpd). Two alternating, 3-horsepower pumps (HP), rated at 160 gallons per minute (gpm), are used to discharge

wastewater into the ponds, where either Pond No. 1 or Pond No. 2 serve as the primary pond. Wastewater gravity drains through overflow risers through the remaining ponds. Ponds No. 1 through 5 were installed in 1962, and Pond No. 6 was installed in 1989. The headworks are at the northwestern corner of the WWTF, immediately west of Pond No. 1.

6. Ponds No. 1 through 6 occupy 12.7 acres. Ponds No. 2, 4 and 5 are about 300 feet square. Pond No. 1 is approximately 400 feet by 250 feet, Pond No. 3 is about 300 feet by 250 feet, and Pond No. 6 is about 600 feet by 350 feet. Ponds No. 1 through 4 concentrically deepen to a maximum of about 9 feet. Pond No. 5 is 8 feet in depth, but has an east-west trending, 2-foot high baffle across the center of the pond. Pond No. 6 is about feet in depth.
7. An irrigation pump (rated at 150 gpm) is used to transfer the wastewater to adjacent land application areas or to operate sprinklers in Ponds No. 1 and 6. Approximately 13 million gallons of pond effluent are discharged to either 3-acres on site or the adjacent land (20-acres) spreading areas annually in order to maintain freeboard in the ponds. The sprinklers in these ponds are used to enhance evaporation and oxidation of the wastewater.
8. The WWTF expansion area will consist of nine additional ponds (Ponds No. 7 to 15) that will occupy 46.7 acres, as shown on Attachment C. The expansion area is immediately east of the South Fork Willow Slough. The Discharger has purchased two 5-HP pumps (rated at 300 gpm) to upgrade the lift station for the proposed development. Each pond in the expansion is estimated to have capacity for wastewater flows from at least 30 equivalent domestic units (EDUs). Ponds No. 7, 8, 10, 11, 13 and 14 are 5 acres in size, Pond No. 9 is the largest at 7.35 acres, and Ponds 12 and 15 are the smallest at 4.69 acres.
9. Currently, 223 homes are slated for development. Initial development for 131 EDUs will be discharged to Pond No. 7 commencing in Spring/Summer 2001. During this initial phase of expansion, approximately 15 acres are designated for land application, including the 7.35 acres designated as Pond No. 9.
10. Installation of Pond No. 8 is proposed as the next phase of development, and is estimated to occur in 2002. On the average, construction of one additional pond and land application area will be installed with the proposed development of 150 EDUs. It is estimated that the expansion should serve development for at least the next 20 years.
11. The Discharger does not plan to beneficially use the proposed expansion area for any purpose other than disposal through land application of wastewater. For each phase of expansion, the land application area will be plowed annually and native vegetation will cut with a flail mower to control excessive growth.
12. The average dry weather flows (ADWF) to the current WWTF are estimated to be 150,000 gallons per day (gpd). Average wet weather flows are estimated to be 180,000 gpd. Flow estimates are based on the hour meters of the pumps at the headworks. The proposed complete expansion of the WWTF will accommodate ADWF of 360,000 gpd, with peak flows as high as

600,000 gpd. Installation of a flow meter was not included for installation in the current or in future expansions of the WWTF.

13. The pond wastewater has most recently (January 2001) been characterized for electrical conductivity (EC) as 830 to 942 micromhos per centimeter ($\mu\text{mhos/cm}$), and pH as 7.6 to 9.2 pH Units. These measurements were collected in excess of the requirements of WDRs Order No. 89-165.
14. Since issuance of WDRs Order No. 89-165, the wastewater ponds have experience operational problems. The insufficient storage capacity of the ponds, and the threatened discharge to surface water drainage courses, have been identified as problems during inspections and in correspondence. Complaint inspections were conducted on 29 September 1992, 13 March 1998, and 24 August 1999. Inspections conducted during March and April 1998 indicated inadequate freeboard, and on 28 April 1998, a Notice of Violation was issued. The Discharger made operational improvements to mitigate the potential spillage of wastewater from the ponds, including the installation of the sprinklers, and an emergency containment pond west of the existing WWTF.
15. Board Staff have previously notified the Discharger of the requirements of Title 22 for the recycling of partially treated wastewater off site from the WWTF. Board staff notified the Discharger of these requirements on 29 September 1992 and again on 27 August 1999. The RWD indicates that the Discharger wishes to continue to use the adjacent Traylor property for land application, but does not want to incur additional Title 22 requirements for the expansion of the WWTF.
16. The potable water supply for the town of Esparto is groundwater. The number and depth of the supply wells are unknown. The Discharger has recently collected a sample (21 November 2000) from Well No. 6, which had an EC of 598 $\mu\text{mhos/cm}$ and a pH of 7.93 pH Units.
17. Three shallow groundwater monitoring wells (W1, W2 and W3) were reportedly installed at the WWTF in 1988 to a depth of 40 feet below ground surface (bgs). All three wells are installed immediately adjacent to the existing wastewater ponds, but have frequently been reported as dry. Groundwater flow direction is unknown, as surveyed well elevations could not be found. The Discharger proposed installation of two additional monitoring wells for the expansion area in the RWD.
18. "Background quality" of the shallow groundwater has not been assessed for this site. The Discharger has been relatively consistent in collecting EC data from the shallow groundwater monitoring wells in compliance with WDRs Order No. 89-165, but nitrate data has not been consistently submitted. The most recent water quality samples, collected in January 2001, reported the following:

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Monitoring Well	Groundwater Depth (feet bgs)	pH (pH Units)	EC (µmhos/cm)
1	29	7.32	531
2	24	7.55	1318
3	23	8.05	919

19. A brief review of the historical shallow groundwater monitoring data indicate that there are consistent differences in concentrations of EC between the existing monitoring wells, which suggests that groundwater may be impacted by releases from the pond. However, these data also suggest that groundwater may be hydraulically connected to the flow in the slough(s).
20. Based on the information provided in previous WDRs, the underlying soils are of the Rincon-Marivn-Tehama association consisting primarily of silty-clay loams with highly variable drainage potential.
21. Surrounding land use is for agriculture. A permanent residence and farmstead are approximately 1800 feet south of the WWTF.
22. As reported in previous WDRs, the annual rainfall is about 16.8 inches with an annual evaporation rate of about 52 inches.
23. The ground surface elevation near the Discharger's wastewater ponds ranges from approximately 170 feet mean sea level (msl) near Pond No. 1 to 166 feet msl near Pond No. 6. The approximate elevation of the bottom of Pond No. 6 is approximately 167 feet msl, Ponds No. 1 and 5 are at about 164 feet msl, Ponds No. 2 and 3 at about 163 feet msl, and Pond No.4 at about 162 feet msl.
24. Surface water drainage in the area of the domestic wastewater ponds easterly with drainage to Lamb Valley Slough and to South Fork Willow Slough. Lamb Valley Slough is tributary to South Fork Willow Slough with the confluence at the southeastern corner of the existing WWTF. South Fork Willow Slough is tributary to the Yolo Bypass. The site is immediately north of the Lower Cache Creek Hydrologic Area Boundary (No. 511.30), as depicted on interagency hydrologic maps prepared by the Department of Water Resources in August 1986.
25. The Board adopted a Water Quality Control Plan, Fourth Edition, for the Sacramento/San Joaquin River Basins (hereafter Basin Plan), which contains water quality objectives for all waters of the Basin. These requirements implement the Basin Plan.
26. The beneficial uses of the Yolo Bypass are agricultural supply for irrigation and stock watering, contact and non-contact recreation, warm freshwater habitat, fish migration and spawning, wildlife habitat and groundwater recharge.
27. The beneficial uses of underlying ground water are domestic, municipal, industrial, process and agricultural supply.

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28. Water quality criteria associated with the protection of surface and/or groundwater beneficial uses are as follows:

<u>Compound</u>	<u>Units</u>	<u>Concentrations¹</u>
BOD ²	mg/l	40/80
pH ³	pH Units	6.5 to 8.5
EC ⁴	µmhos/cm	700 ⁵ , 900, 1600, 2200
TDS ⁴	mg/l	450 ⁵ , 500, 1000, 1500
Nitrate ⁶	mg/l	10/45

BOD = Biochemical Oxygen Demand (5-day)

TDS = Total Dissolved Solids

¹ Source = *A Compilation of Water Quality Goals*, California Environmental Protection Agency, Regional Water Quality Control Board, Central Valley Region, August 2000.

² Monthly average and daily maximum concentrations, respectively. Technological standards easily achieved with proper design and operation.

³ Secondary Maximum Contaminant Level

⁴ Recommended level, Upper-level, and Short-term level, respectively for Secondary Maximum Contaminant Level (MCL).

⁵ Agricultural Water Quality Goal

⁶ Primary MCL, as Nitrogen and as Nitrate, respectively.

29. Water quality criteria associated with the protection of groundwater beneficial uses will be determined during the Groundwater Assessment Report required by this Order.
30. The Board has considered antidegradation pursuant to State Board Resolution No. 68-16 and finds that insufficient data exists to determine whether this discharge is consistent with the provisions of the Resolution. Therefore, the requirements of this Order and the Monitoring and Reporting Program (MRP) include groundwater monitoring to determine background groundwater quality and whether the discharge will cause an increase above background levels. If the discharge causes an increase, the Discharger may be required to mitigate the impacts of the discharge through such actions as source control, alternative disposal methods, cessation of discharge, or any combination of these or other alternatives, that will prevent groundwater degradation.
31. The California Department of Health Services (DHS) has recommended that facilities storm water discharges from wastewater recycling areas be fully contained and returned year round to storage ponds. Storm water runoff from these areas can be allowed provided the following conditions are met:
- Wastewater applied to the land application area is disinfected year round according to DHS' *Uniform Guidelines for the Disinfection of Wastewater*.
 - Disinfection of the wastewater to a 23 most probable number per hundred milliliters (MPN/100 ml) monthly median with a 240 MPN/100 ml daily maximum is to be maintained.

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- c. Wastewater application is ceased a minimum of 48 hours prior to a diversion of storm water event to the adjacent surface water body.
 - d. During wastewater application, all tailwater and/or storm water is returned to the wastewater ponds.
 - e. All wastewater is absorbed into the soil of the application area, tailwater collection and return system, or returned to the wastewater ponds within 24 hours after application has ceased.
 - f. No standing water is allowed at any time in the tailwater collection and return system.
 - g. Any commingled wastewater and storm water must be contained and returned to the wastewater ponds.
 - h. Prior to release, storm water runoff from the land application area shall be checked for chlorine concentration.
32. The Board reviewed the 21 January 1999 Negative Declaration of Environmental Impact for the Conditional Use Permit for the expansion of the existing WWTF for the Town of Esparto, and certified by the Yolo County Planning Commission on 11 February 1999. The County found that impacts from the project would be less-than-significant because it would not violate existing standards.
33. On 30 December 1998, Board staff responded to the Request for Comments for the Initial Study from the Yolo County Planning and Public Works Department, regarding the expansion of the existing WWTF for the Town of Esparto. Comments stated that the WDRs for this site would be revised, surface water discharge was prohibited, and the Regional Board would require that groundwater monitoring. Compliance with the Provisions of this Order will assure that less-than-significant impacts will occur due to the expansion of the WWTF.
34. The action to update waste discharge requirements for this facility is exempt from the provisions of the California Environmental Quality Act (CEQA), in accordance with California Code of Regulations (CCR), Title 14, Section No. 15261. This exemption reflects the ongoing nature of the operator's wastewater facilities project, originally approved in 1963.
35. This discharge is exempt from the requirements of Consolidated Regulations for Treatment, Storage, Processing, or Disposal of Solid Waste, as set forth in Title 27, California Code of Regulations (CCR), Division 2, Subdivision 1, Section 20005, et seq., (hereinafter Title 27). The exemption, pursuant to Section 20090(b), is based on the following:
- a. The Board is issuing waste discharge requirements, and
 - b. The discharge complies with the Basin Plan, and
 - c. The wastewater does not need to be managed according to 22 CCR, Division 4.5, Chapter 11, as a hazardous waste.
36. Section 13267(b) of California Water Code provides that: "In conducting an investigation specified in subdivision (a), the regional board may require that any person who has discharged, discharges, or is suspected of discharging, or who proposes to discharge within its region, or any

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citizen or domiciliary, or political agency or entity of this state who has discharged, discharges, or is suspected of discharging, or who proposes to discharge waste outside of its region that could affect the quality of the waters of the state within its region shall furnish, under penalty of perjury, technical or monitoring program reports which the board requires. The burden, including costs of these reports, shall bear a reasonable relationship to the need for the reports and the benefits to be obtained from the reports."

37. The Board has notified the Discharger and interested agencies and persons of its intent to prescribe waste discharge requirements for this discharge and has provided them with an opportunity for a public hearing and an opportunity to submit their written views and recommendations.
38. The Board, in a public meeting, heard and considered all comments pertaining to the discharge.

IT IS HEREBY ORDERED that Order No. 89-165 is rescinded and the Yolo County Housing Authority and the California Department of Housing and Community Development, their agents, successors, and assigns, in order to meet the provisions contained in Division 7 of the California Water Code and regulations adopted thereunder, shall comply with the following:

A. Discharge Prohibitions:

1. Discharge of wastes to surface waters or surface water drainage courses is prohibited.
2. Bypass or overflow of untreated or partially treated waste is prohibited, except as discussed in the Standard Provisions.
3. Discharge of waste classified as 'hazardous,' defined in Section 20164 of Title 27, CCR, or 'designated', as defined in Section 13173 of the California Water Code, is prohibited.
4. The discharge of wastewater from the ponds to the land application area is prohibited during the period extending from 1 November to 1 April.
5. The discharge of irrigation tail waters beyond the boundaries of the land application area, which is designated for wastewater disposal, is prohibited.

B. Discharge Specifications:

1. Neither the treatment nor the discharge shall cause a nuisance or condition of pollution as defined in the California Water Code, Section 13050.
2. The discharge shall not cause degradation of any water supply.

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3. The monthly ADWF from the WWTF shall not exceed 360,000 gpd. The maximum wet weather flow shall not exceed 600,000 gpd.
4. The treatment and disposal facilities shall be operated and maintained to prevent inundation or washout due to floods with a 100-year return frequency.
5. The wastewater ponds shall have sufficient capacity to accommodate allowable wastewater, and design seasonal precipitation and ancillary inflow and infiltration at all times.
6. Objectionable odors originating at this facility shall not be perceivable beyond the limits of the property owned by the Discharger.
7. As a means of discerning compliance with Discharge Specification No. 6, the dissolved oxygen content in the upper zone (1 foot) of the wastewater ponds shall not be less than 1.0 milligram per liter (mg/l).
8. The wastewater ponds shall not have an average daily pH of less than 6.5 or greater than 9.0. Revision of this limitation may be necessary after characterization of the source water as documented in the Groundwater Assessment Report, as required by Provision 6.E. of this Order.
9. The wastewater ponds shall be managed to prevent breeding of mosquitoes. In particular,
 - a. An erosion control program shall assure that small coves and irregularities are not created around the perimeter of the water surface.
 - b. Weeds shall be minimized through control of water depth, harvesting, or herbicides.
 - c. Dead algae, vegetation, and debris shall not accumulate on the water surface.
10. A minimum freeboard of 2.0 feet (measured vertically to the lowest point of the berm) shall be maintained in the ponds at all times.
11. Public contact with wastewater in the wastewater ponds shall be precluded through such means as fencing, signs and other acceptable alternatives.

C. Land Application Area Specifications:

1. The use of water from the wastewater ponds shall be for the land application area only.
2. Wastewater discharged to the land application area shall not exceed the following average limits:

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<u>Constituent</u>	<u>Units</u>	<u>Limit</u>
pH	pH Units	6.5 to 9.0
Total Coliform	MPN/100 ml	23 ¹ , 240 ²

¹ Monthly Median

² Daily Maximum

3. Treated wastewater shall remain within the land application area at all times.
4. Controlled buffer zones shall be maintained around the land application area during wastewater application, with setbacks for fallow land based on the following criteria.

<u>Setback Distance (feet)</u>	<u>Site Condition</u>
25	Property Boundary
30	Roads
50	Surface Water Drainage Course
150	Water Well

5. Treated wastewater used for evaporation/percolation shall be managed to minimize erosion.
6. Application of the treated wastewater to the land application area shall be managed to minimize percolation to groundwater.
7. There shall be no visible standing water in the land application area for more than 48 hours in any consecutive 7-day period after application of the wastewater.
8. Irrigation of the land application area shall be managed to prevent breeding of mosquitoes, where:
 - a. Ditches are maintained to be free of emergent, marginal, and floating vegetation; and
 - b. Low-pressure and unpressurized pipelines and ditches accessible to mosquitoes shall not be used to store water.
9. The user shall not discharge to the land application area during periods of rainfall or when the ground is saturated.
10. Public contact with wastewater in the land application area shall be precluded through such means as fencing, signs and other acceptable alternatives.

D. Sludge Disposal:

1. Collected screenings, sludges, and other solids removed from liquid wastes shall be disposed of in a manner that is consistent with Title 27, Division 2, Subdivision 1 of the CCR and approved by the Executive Officer.
2. Use and disposal of sewage shall comply with existing Federal and State laws and regulations, including permitting requirements and technical standards included in the Code of Federal Regulations (CFR) 40 CFR 503.

If the State Water Resources Control Board and the Regional Water Quality Control Boards are given the authority to implement regulations contained in 40 CFR 503, this Order may be reopened to incorporate appropriate time schedules and technical standards. The Discharger must comply with the standards and time schedules contained in 40 CFR 503 whether or not they have been incorporated into this Order.

3. Any proposed change in sludge disposal from a previously approved practice (as described in these WDRs) shall be reported to this office at least 90 days in advance of the change.

E. Ground Water Limitations:

The discharge, in combination with other sources, shall not cause underlying groundwater to contain waste constituents in concentrations statistically greater than background water quality, except for coliform. For coliform, increases shall not cause the most probable number of total coliform organisms to exceed a median value of 2.2/100 ml over any 7- day period.

F. Provisions:

1. The Discharger shall comply with the MRP No. 5-01-112, which is part of this Order, and any revisions thereto as ordered by the Executive Officer.
2. The Discharger shall comply with the "Standard Provisions and Reporting Requirements for Waste Discharge Requirements", dated 1 March 1991, which are attached hereto and by reference a part of this Order. This attachment and its individual paragraphs are commonly referenced as "Standard Provision(s)."
3. In the event of any change in control or ownership of land or waste discharge facilities described herein, the Discharger shall notify the succeeding owner or operator of the existence of this Order by letter, a copy of which shall be immediately forwarded to this office.
4. At least 90 days prior to termination or expiration of any lease, contract, or agreement involving disposal or reclamation areas or off-site reuse of effluent, used to justify the capacity authorized herein and assure compliance with this Order, the Discharger shall

notify the Board in writing of the situation and of what measures have been taken or are being taken to assure full compliance with this Order.

5. The Discharger must comply with all conditions of this Order, including timely submittal of technical and monitoring reports as directed by the Executive Officer. Violations may result in enforcement action, including Regional Board or court orders requiring corrective action or imposing civil monetary liability, or in revision or rescission of this Order.
6. The Discharger shall submit the following documents according to the following time schedule to assure compliance with this Order:

<u>Task</u>	<u>Compliance Date</u>
A. Flow Meter Installation	1 August 2001
B. Land Application Baseline Soil Sampling Report	1 August 2001

Baseline sampling shall be conducted in the WWTF expansion area to assess the concentrations of salinity, nitrogen, phosphorus and potassium in the soil prior to application of excess domestic wastewater. A minimum of two composite soil samples shall be collected from the land application area. Each composite sample shall be comprised of four to ten discrete soil samples collected throughout the land application area. These data shall be reported in tabular form. Baseline soil data shall be used to compare to annual land application data to assess whether wastewater has been applied in excess of the nutrient needs of the native vegetation.

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| C. Monitoring Well Installation Workplan | 1 August 2001 |
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Prior to construction, plans and specifications for installation of a groundwater monitoring wells shall be submitted to Board staff for review and approval. New monitoring wells shall be installed to monitor groundwater flow direction and quality in the area of the WWTF expansion. The new monitoring wells shall be used to assess background concentrations in shallow groundwater and to assist with the determination of whether there is a potential impact from the wastewater ponds and land application area, if any, on the shallow groundwater. The data shall also be used to evaluate potential impacts from other sources. The workplan shall include the following:

- location of the well on a map of the site;
- drilling and development methods;
- well details as shown on a construction diagram;
- schedule of field activities.

D. Well Installation Report

1 November 2001

Upon installation of the monitoring wells, a report of the field activities shall be submitted. The report shall include driller's logs and as-builts of the well construction. The Report shall include the analytical results of the initial sampling event from each newly installed monitoring well. The initial sampling events from these newly installed wells shall serve as an indication of "background" groundwater quality. Sampling event may be conducted independently or as part of the required monitoring under this Order for the compounds specified in MRP No. 5-01-112. Analytical results shall be presented in tabular form.

E. Groundwater Assessment Report

1 February 2003

The purpose of the Groundwater Assessment Report shall be to evaluate potential impacts from the existing WWTF to assure that continued expansion of the facility does not degrade groundwater quality. The Discharger shall consider the water quality of the source water supply, groundwater quality data from the existing monitoring wells, and background groundwater quality. Shallow background groundwater quality is to be collected from the newly installed monitoring wells (as required under Provisions 6.A and B of this Order) and shall be based, at a minimum, on four consecutive quarters to detect potential seasonal variations within the data set.

The evaluation for the source water characterization may be obtained from supply well samples collected specifically for this assessment or from other sampling venues such as source water assessment of the supply wells, typically conducted at the time of permitting by the Yolo County Environmental Health Department. However, if data collected from outside sources was collected prior to 2001, the Discharger must supply at least two additional measurements for pH and EC, collected during different seasons, to validate the existing data.

The Discharger shall provide a determination of whether shallow groundwater has been impacted by the operation of the WWTF. This determination shall be made, in part, on a comparison of up and downgradient groundwater quality data, considering potential influence from surface water due to the potential hydraulic connection between the wells and the slough.

All analytical results for the Report shall be presented in tabular form. Groundwater elevation maps depicting the flow direction and magnitude and representative of the wet and dry seasons shall be presented in the Report. The Discharger shall also provide hydrographs and trend graphs for the three existing monitoring wells (W1, W2 and W3) for groundwater elevations and EC concentrations, respectively, over the historical record. Trend graphs may also be prepared for nitrate, provided sufficient data are available.

F. Site Mitigation Plan

1 August 2003

In the event the assessment of groundwater indicates that degradation has occurred, the Discharger shall provide a plan describing the measures to be taken to mitigate further releases from the WWTF. The plan shall include the following information, at a minimum:

- Description of wastewater collection, treatment and disposal system, as modified;
- Design specifications for new facilities required, including capacity and hydraulic mass balance (1-in-100 year return frequency) if a pond system is installed;
- Construction methods and materials;
- Schedule for implementation;
- Operation and maintenance plan(s); and
- Proposals for monitoring modifications, as appropriate.

G. Phase Design Plan

**3 Months Prior to
Construction**

The Discharger shall submit a design plan for each proposed phase of WWTF expansion, where an additional pond is to be constructed to accommodate development. This plan shall identify the design EDU for that phase, and shall contain the proposed upgrades to the headworks; pond specifications (including dimensions, elevations, piping and valving); and a hydraulic balance for the 100-year annual return for the entire pond system. The plan shall also identify the estimated ADWF, AWWF, and the associated land spreading area for that phase of development. A schedule for construction and commencement of operation shall also be provided.

H. Revised Operation and Maintenance Plan

**60 days after Phase
Construction**

The Discharger shall submit a plan that describes the methods and procedures to be used for operation of the WWTF, including the pond system and land application area, with each phase of expansion. The plan shall address the monitoring and operations needed to assure that sufficient hydraulic capacity is available in the ponds at all times for protection from the 100-year flood event. The plan shall outline measures to be taken should freeboard specifications be threaten or violated. These measures shall include, but not be limited to, off-site disposal of wastewater at approved and permitted wastewater treatment facility. Off-site disposal alternatives shall be supported by "will serve" letter(s) from an approved and permitted facility.

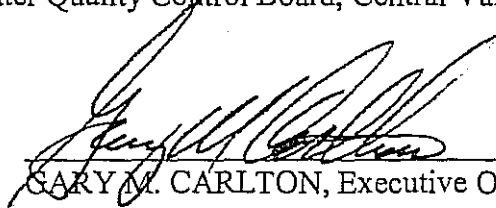
7. A copy of this Order shall be kept at the discharge facility for reference by operating personnel. Key operating personnel shall be familiar with its contents.

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7. A copy of this Order shall be kept at the discharge facility for reference by operating personnel. Key operating personnel shall be familiar with its contents.
8. The Board will review this Order periodically and will revise requirements when necessary.

I, GARY M. CARLTON, Executive Officer, do hereby certify the foregoing is a full, true, and correct copy of an Order adopted by the California Regional Water Quality Control Board, Central Valley Region, on 11 May 2001.



GARY M. CARLTON, Executive Officer

CKW/11 May 2001

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

MONITORING AND REPORTING PROGRAM NO. 5-01-112

FOR
ESPARTO WASTEWATER TREATMENT FACILITY

ESPARTO COMMUNITY SERVICES DISTRICT
YOLO COUNTY

This Monitoring and Reporting Program (MRP) describes requirements for monitoring surface water, wastewater, land application area, and groundwater. This MRP is issued pursuant to Water Code Section 13267. The Discharger shall not implement any changes to this MRP unless and until a revised MRP is issued by the Executive Officer.

MONITORING

All samples shall be representative of the volume and nature of the discharge or of the matrix of the material sampled. Time, date, and location of sample collection shall be recorded and presented in the data summary for the monitoring period.

Analyses for the following compounds are to be performed using the following Environmental Protection Agency Methods, Standard Methods, or equivalent methods:

<u>Compound</u>	<u>Method</u>
pH	150.1 and/or Field Meter
Electrical Conductivity (EC)	120.1 and/or Field Meter
Dissolved Oxygen (DO)	360.1 and/or Field Meter
Chlorine Residual	4500 and/or Field Meter
Biochemical Oxygen Demand (BOD) _{5-day}	5210B, 405.1
Total Suspended Solids (TSS)	2540D, 160.2
Total Coliform	909A
Nitrates (as Nitrogen)	300, 352.1
Total Dissolved Solids (TDS)	2540C, 160.1
General Minerals ¹	as above, and 200.7, 300.0, 2310B, 2340B

¹ General Minerals shall include at least the following compounds: calcium, potassium, sodium, chloride, nitrate, sulfate, total dissolved solids, electrical conductivity, pH, total alkalinity, and total hardness.

SURFACE WATER MONITORING

Surface water samples shall be collected from Lamb Valley Slough and South Fork Willow Slough. These samples are needed to assess whether groundwater quality data is influenced by the presence of flow in the slough(s) and/or impacts from the wastewater ponds. Surface water samples shall be

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collected at three points, R1, R2, and R3 (as designated on Attachment C), when flowing, for the following:

<u>Constituents</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Frequency</u> ¹
Flow	Yes/No	Observation	Monthly
pH	pH Units	Grab ²	Quarterly
EC	µmhos/cm	Grab ²	Quarterly

¹ The Discharger may propose a reduction in frequency to semi-annual provided the data demonstrate no significant changes in water quality. Requests for reduction in monitoring must address the requirements of the Monitoring Reduction Request Report, as defined under the Reporting section below. Reductions in monitoring requirements shall not occur until approved by the Executive Officer.

² Samples shall be collected when there is flow in the slough and shall be collected contemporaneously with the groundwater monitoring event.

WASTEWATER PONDS

Samples shall be collected from each of the wastewater ponds in operation at the time of sampling for sample analysis. Wastewater pond monitoring shall include the following:

<u>Constituents</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Frequency</u> ¹
Flow	Gallons/day	Influent Estimate	Cumulative
Freeboard	Feet	Measurement	Weekly
pH ²	pH Units	Grab	Monthly
EC	µmhos/cm	Grab	Monthly
DO ²	mg/l	Grab	Monthly
BOD _{5-day}	mg/l	Grab	Monthly
TSS	mg/l	Grab	Monthly
Nitrates (as Nitrogen)	mg/l	Grab	Monthly
TDS	mg/l	Grab	Semi-Annually ³

¹ The monitoring frequency may be reduced upon written request from the Discharger and supported by proven performance data. At a minimum, the primary and finishing ponds must be monitored quarterly. Requests for reduction in monitoring must address the requirements of the Monitoring Reduction Request Report, as defined under the Reporting section below. Reductions in monitoring requirements shall not occur until approved by the Executive Officer.

² If the compound is outside the permissible limits established in Discharge Specification Nos. B.7 and B.8 of Order No. 5-00-112, the Discharger shall resample within the designated sampling frequency period to determine if compliance was achieved for the monitoring period.

³ Semi-annual monitoring shall be conducted at the height of the dry and wet seasons (typically August/September and January/February, respectively).

LAND APPLICATION AREA MONITORING

The Discharger shall monitor the effluent discharged to, soil within, and storm water releases from the land application area. Effluent monitoring is needed assure adequate disinfection of the wastewater. Soil monitoring is needed to ascertain whether the nitrate-nitrogen loading is in excess of that required by the vegetation. A minimum of two composite soil samples shall be collected from the land application area. Each composite sample shall be comprised of four to ten discrete soil samples collected throughout the land application area. Runoff from the land application area is needed to assure that storm water releases do not convey pollutants to surface water bodies. Samples shall be analyzed for the following compounds:

Effluent

<u>Constituents</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Frequency</u> ¹
Flow	Gallons/day	Continuous	Monthly
pH	pH Units	Grab	Monthly
EC	µmhos/cm	Grab	Monthly
Total Coliform	MPN/100' ml	Grab	Monthly

¹ During periods of land application only.

Soil

<u>Constituents</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Frequency</u> ¹
Salinity	mg/kg	Composite	Annual
Nitrogen	mg/kg	Composite	Annual
Phosphorus	mg/kg	Composite	Annual
Potassium	mg/kg	Composite	Annual

¹ Samples shall be collected prior to the onset of the rainy season for only those periods in which discharge occurs.

Storm Water Releases

<u>Constituents</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Frequency</u> ¹
pH	pH Units	Grab	2 Events
EC	µmhos/cm	Grab	2 Events
Total Coliform	MPN/100 ml	Grab	2 Events
Chlorine Residual	mg/l	Grab	2 Events

¹ Samples shall be collected prior to the first release of storm water from the application area.

GROUNDWATER

Monitoring wells designated for groundwater monitoring of the wastewater ponds shall be installed with the approval of Board staff. Groundwater samples shall be collected for analysis from each well after the well has been adequately purged. Purging is typically defined as either three well volumes, until stabilization of temperature, pH and EC, or refilling of the well after dewatering. Each groundwater monitoring well shall be monitored for the following:

<u>Constituents</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Frequency^{1, 2}</u>
Groundwater Elevation ³	Feet	Measurement	Quarterly
pH	pH Units	Grab	Quarterly
EC	µmhos/cm	Grab	Quarterly
Nitrates (as Nitrogen)	mg/l	Grab	Quarterly
TDS	mg/l	Grab	Quarterly
Total Coliform Organisms	MPN/100ml	Grab	Quarterly
General Minerals	mg/l	Grab	Semi-Annually ^{2, 4}

- ¹ The Discharger may propose a reduction in frequency to semi-annual provided the data demonstrate no significant changes in water quality. Requests for reduction in monitoring must address the requirements of the Monitoring Reduction Request Report, as defined under the Reporting section below. Reductions in monitoring requirements shall not occur until approved by the Executive Officer.
- ² Semi-annual monitoring shall be performed at the height of the dry (typically August/September) and wet (typically January/February) seasons.
- ³ Groundwater elevation shall be determined based on depth-to-water measurements from a surveyed measuring point elevation on the well.
- ⁴ A "fingerprint" of the groundwater, for characterization purposes, shall be conducted for two years.

OBSERVATIONS

Each monitoring report shall include observations of the general condition of the wastewater treatment facility. These observations may be kept in a log and shall include the following:

1. Odors;
2. Wastewater color;
3. Levee conditions, including rodent holes, piping, and bank erosion;
4. Vegetative growth in and around the ponds and land application area;
5. Condition of fencing and signs;
6. Dates of changes in pond operation, such as change in primary pond designation;
7. Dates, occurrences, location and estimated amounts of leaks and spillage from the ponds or application area;

8. Dates of routine maintenance associated with control of vegetation around the ponds and land application area; and with transferal of the with drying of the ponds and or removal of solids/sludges; and
9. Dates and descriptions of maintenance activities associated with levee repair or control of septic conditions and unauthorized releases of wastewater from the ponds or application area.

REPORTING

In reporting the monitoring data, the Discharger shall arrange the data in tabular form so that the date, the constituents, and the concentrations are readily discernible. The type of sample (composite or grab) must be identified. The results of any monitoring done more frequently than required at the locations specified in the Monitoring and Reporting Program shall be reported to the Board. All monitoring reports shall include chain-of-custody forms and analytical laboratory data sheets. Time, date, and location of sample collection shall be recorded and presented in the data summary for the monitoring period, consistent with the supporting analytical data sheet.

A. Monthly Reports

Monthly Reports for pond monitoring shall be submitted to the Regional Board by the first day of the second month following the month of sampling (i.e., the January monthly report is due by **1 March**). Two monthly monitoring reports are to be submitted per calendar quarter. At a minimum, the reports shall include the following:

1. A scaled map showing relevant structures and features of the facility and the locations of all sampling stations;
2. The analytical results of all wastewater pond, and land application area monitoring, and observations logs and records of sludge, septage, and/or grease removed from the wastewater treatment and disposal facilities;
3. A comparison of the monitoring data to the discharge specifications and an explanation of any violation of these requirements.

B. Quarterly Reports

Quarterly Monitoring Reports shall be submitted to the Regional Board by the **1st day of February** (4th Quarter Report), **May** (1st Quarter Report), **August** (2nd Quarter Report), and **November** (3rd Quarter Report). At a minimum, the report shall contain:

1. A scaled map showing relevant structures and features of the facility and the locations of all sampling stations, including surface water stations and groundwater monitoring wells;

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2. All of the analytical results representative of a third monthly monitoring event for the current calendar quarter. These data are generated as a part of the monthly monitoring requirements for the wastewater ponds and land application area, as specified under **A. Monthly Reports**, above;
3. Analytical results for quarterly monitoring conducted for surface water and groundwater;
4. Analytical results of storm water releases from the land application area shall be summarized in the Second Quarter monitoring report.
5. A comparison of monitoring data to the discharge specifications, effluent limitations, and water quality criteria protective of beneficial uses as identified in this Order or as determined in the Groundwater Assessment Report (required by this Order). An explanation of possible causes shall be provided when the analytical results exceed the specification, limitation, or criterion for a specific compound;
6. Calculation of groundwater elevations and determination of the groundwater flow direction and gradient on the date of measurement. A groundwater flow map shall be provided, which depicts the groundwater elevations in each monitoring well, and the direction and magnitude of flow;
7. A narrative discussion of the analytical results for all media and locations monitored, including spatial and temporal trends, with reference to summary data tables, graphs, and appended analytical reports (as applicable). Any changes in surface water and/or groundwater flow and magnitude from previous monitoring events shall be identified; and
8. A narrative description of all preparatory, monitoring, sampling, and analytical testing activities. The narrative shall be sufficiently detailed to verify compliance with the WDRs, this MRP, and the Standard Provisions and Reporting Requirements. The narrative shall be supported by field logs for each well documenting depth to groundwater; parameters measured before, during, and after purging; method of purging; calculation of the casing volume; and total volume of water purged.

C. Annual Monitoring Report

An Annual Monitoring Report shall be submitted by **1 February of each year** and may be substituted for the Fourth Quarter Monitoring Report. At a minimum, the Annual Monitoring Report shall include the following:

1. A scaled map showing relevant structures and features of the facility and the locations of all sampling stations;

2. Analytical results of the soil samples collected annually from the land application;
3. A summary of the all the Quarterly Monitoring Reports;
4. A narrative discussion of the analytical results for all media and locations monitored, including spatial and temporal trends during the previous calendar year, with reference to summary data tables, graphs, and appended analytical reports (as applicable). Trend graphs may be prepared to demonstrate static or dynamic changes in the data set. Any changes in surface water and/or groundwater flow and magnitude during the calendar year shall be identified;
5. Information on the disposal of biosolids;
6. A discussion of compliance and the corrective action taken, as well as any planned or proposed actions needed to bring the discharge into full compliance with the waste discharge requirements; and
7. A discussion of any data gaps and potential deficiencies, redundancies and/or excesses in the monitoring system or reporting program.

D. Monitoring Reduction Request Report

In the event the Discharger believes that a reduction in the monitoring requirements of this Order are appropriate, a summary of the analytical results and any other required measurements must be submitted to justify the proposed changes. The purpose of this report is to identify analytical results or other measurements that are redundant, excessive, or represent steady-state conditions, thereby streamlining the monitoring, reporting and review process, and releasing resources for any needed waste management improvements. Board staff will review the report to concur or refute and change the proposed recommendations for changes in the monitoring program. Reductions in monitoring requirements shall not occur until approved by the Executive Officer. At a minimum a monitoring reduction report shall contain the following:

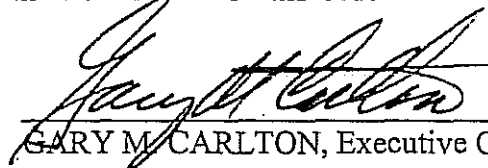
1. A scaled map showing relevant structures and features of the facility and the locations and any other sampling stations;
2. Tabular summaries for the historical period or for the segment of the relevant data record;
3. Trend graphs for the historical period or for the segment of the relevant data record; and
4. Any maps, particularly for groundwater (i.e. groundwater elevation flow maps), which will assist in the demonstration of the consistency of the data record.

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The Discharger shall implement the above monitoring program as of the date of this Order.

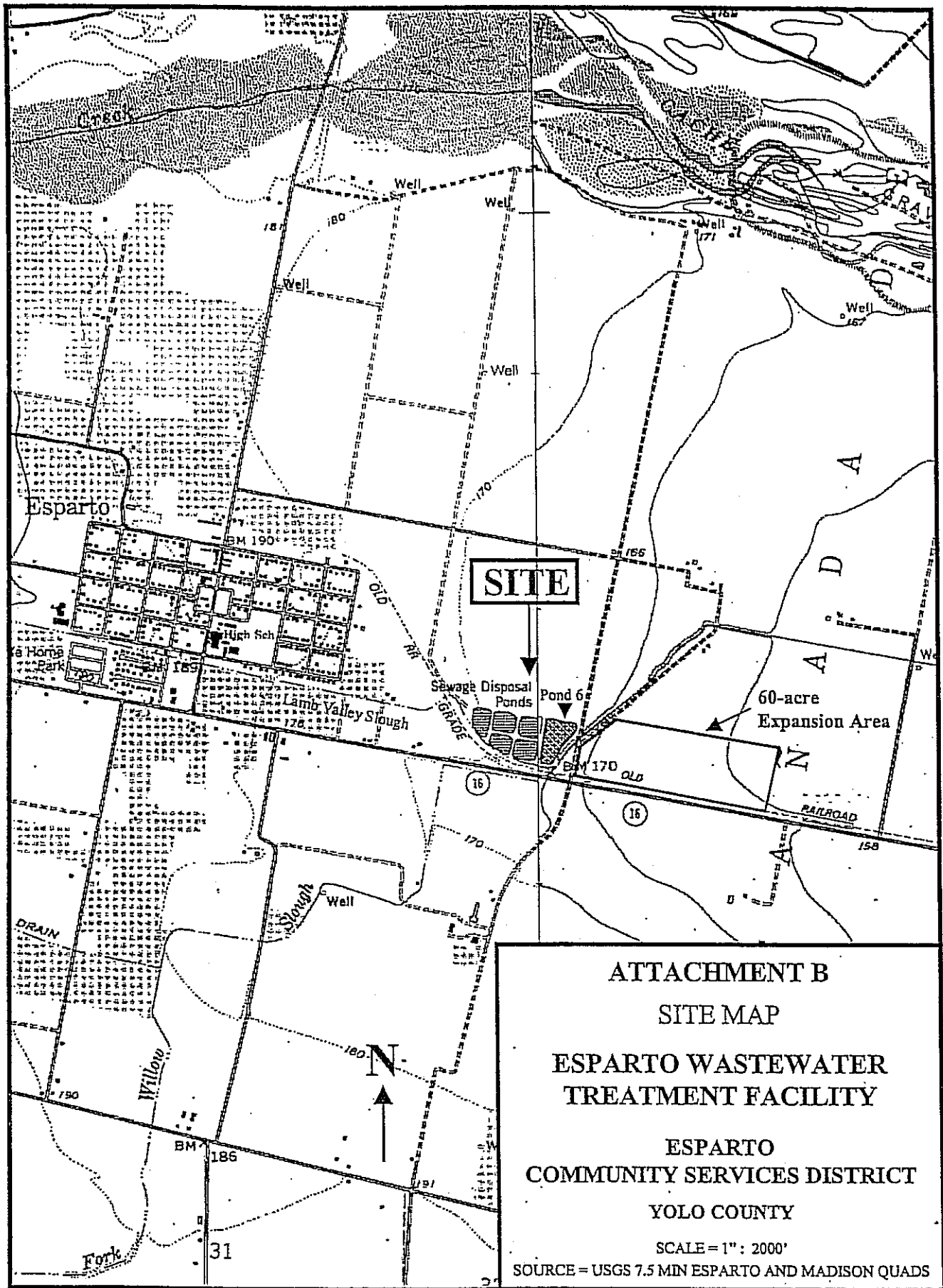
Ordered by :


GARY M. CARLTON, Executive Officer

11 May 2001

(Date)

CKW



ATTACHMENT C

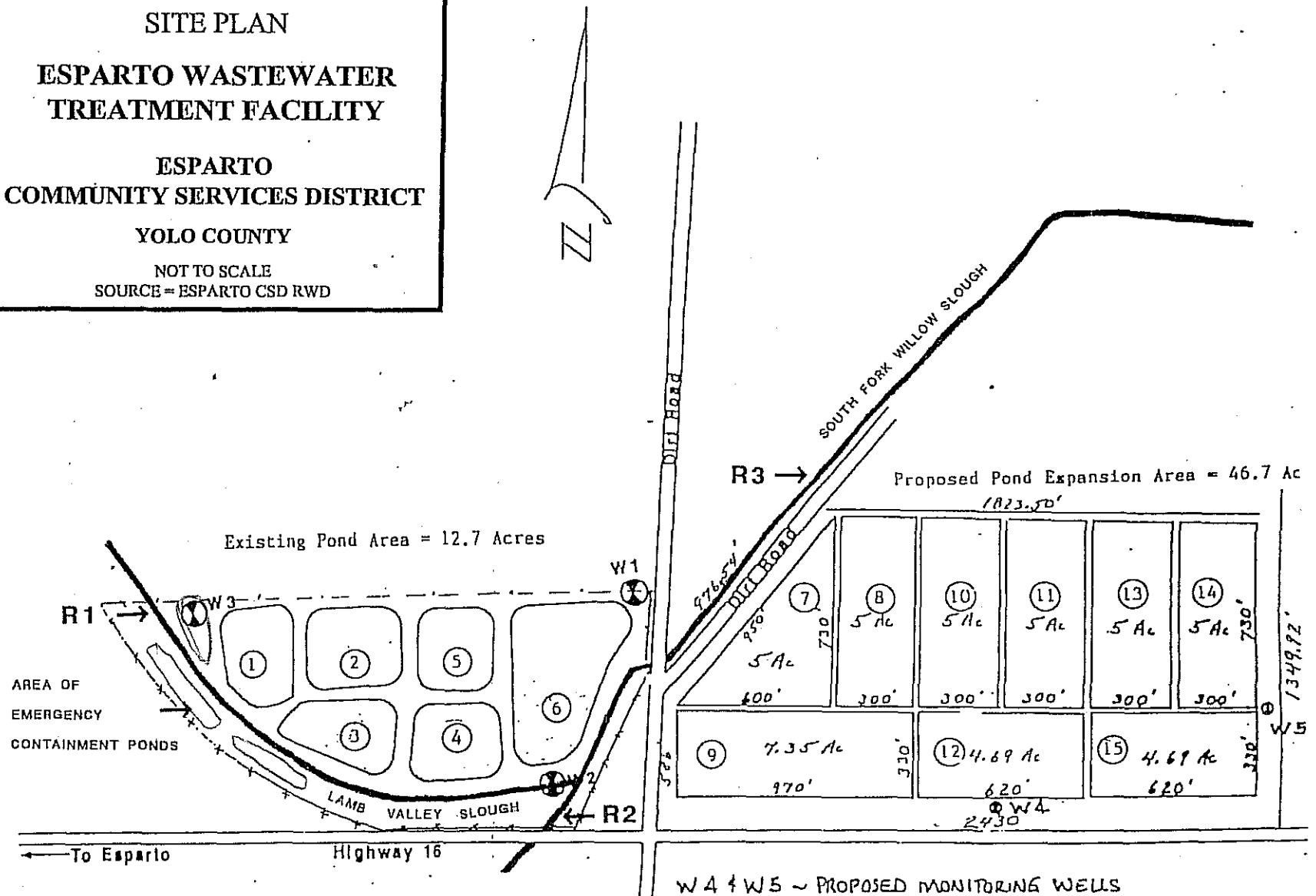
SITE PLAN

ESPARTO WASTEWATER TREATMENT FACILITY

ESPARTO
COMMUNITY SERVICES DISTRICT

YOLO COUNTY

NOT TO SCALE
SOURCE = ESPARTO CSD RWD



INFORMATION SHEET

ESPARTO WASTEWATER TREATMENT FACILITY, ORDER NO. 5-01-112
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The Esparto wastewater treatment facility (WWTF) is owned and operated by the Esparto Community Services District (CSD). WWTF is on a 25-acre parcel, approximately 0.5 miles east of Esparto in Yolo County. In April 2000, the Esparto CSD submitted a Report of Waste Discharge for the expansion of the WWTF. This facility is regulated by Waste Discharge Requirements (WDRs) Order No. 89-165, which are inadequate for regulating the proposed expansion of the WWTF. The WWTF is bordered on the south by the Lamb Valley Slough and on the east by the South Fork Willow Slough.

The Esparto WWTF serves a few dozen business, public schools, and 540 homes, and consists of a wet well/lift station, communitor and six oxidation ponds (Nos. 1 through 6). Pond No. 1 or 2 serves as the primary pond. Average dry weather flows (ADWF) are estimated to be approximately 150,000 gallons per day (gpd), with average wet weather flows (AWWF) of 180,000 gpd. Since issuance of WDRs Order No. 89-165, operational problems associated with pond capacity and potential spillage to Lamb Valley Slough have been reported.

WWTF expansion is proposed to occur in phases. The Esparto CSD purchased 60 acres immediately east of the existing site in February 2000 to accommodate development during the next 20 years. Full build-out of the WWTF will consist of nine additional ponds, each approximately 5 acres in size, and occupying a total of about 47 acres. Estimated ADWF for complete expansion of the WWTF is 360,000 gpd with AWWF of 600,000 gpd.

The water supply for the town of Esparto is groundwater, which has initially been characterized as slightly alkaline. There are three groundwater monitoring wells at the site, and sampling results are also alkaline and suggest potential impacts from the WWTF and a hydraulic connection with the slough(s).

Under these WDRs, wastewater applied to the land application area will be disinfected and application will be during the wet season. Monitoring of the effluent and storm water releases is required to assure adequate disinfection of the wastewater. Installation of monitoring wells in the area of the proposed expansion and groundwater characterization are also required. Surface water sampling is required to assess impacts on groundwater. Monitoring data will be used to determine whether wastewater treatment and disposal has the potential to impact groundwater quality. In addition, these WDRs will require that a design plan and operation and maintenance (O&M) plan be submitted for each phase of expansion. A hydraulic balance for the 100-year annual return and identification of land spreading areas are required in the design plan. Requirements of the O&M plan include identification of measures to be taken to assure adequate hydraulic capacity of the pond system, and actions to be taken pond spillage is threatened or occurs.

The beneficial uses of underlying ground water are domestic, municipal, industrial, and agricultural supply. The beneficial uses of surface water in the Yolo Bypass are agricultural supply for irrigation and stock watering, contact and non-contact recreation, warm freshwater habitat, fish migration and spawning, wildlife habitat and groundwater recharge.

Surface water drainage from the site is to Lamb Valley Slough, tributary to South Fork Willow Slough, tributary to the Yolo Bypass.